

ANDREW THOMAS

andrew.thomas2614@gmail.com • (613) 799-7800 • Kanata, Ontario, Canada
[LinkedIn](#)

SUMMARY

Mechatronics Engineering student with industry experience in embedded C/C++ and Python development for hardware systems, supported by coursework and projects in robotic design. Proven ability to debug complex issues in Linux-based firmware environments, collaborate across HW/SW teams, and deliver features from design through testing. Proactive, driven, and motivated to make an impact with key systems. Seeking 12-16 internships beginning May 2027.

RELATED EXPERIENCE

Robotics Software Development Intern, Provectus Rheinmetall

May 2026 - Aug 2026

- Advanced the autonomy of unmanned ground vehicles through C++ software development
- Designed, implemented, and tested C++ object-oriented solutions for control and planning tasks with industry standard SDLC git processes. Reduced average radar offset calibration time by 80% by adding features to a VTK/OpenCV based visualization utility. Interfaced with a range of exteroception devices including LiDAR, radar, IMU, and GNSS GPS to improve localization through EKF sensor fusion
- Conducted independent field testing at AreaX.O requiring debugging of both hardware and software components.

Embedded Software Engineering Intern, Ciena

May 2025 - Aug 2025

- Completed dozens of Agile-style tickets for a line of photonic traffic cards, ranging from writing and testing REST-conforming microservice functions (e.g. IP checking), to Linux boot-processes (e.g. bind-mounting external disk FS's, post-pivot hostname verification), to CPU performance optimization, all to support a major software release for global clients
- Independently built out a Python/Bash scripted log parsing and analysis tool used across teams (accessible to ~200 SW R&D employees) for new feature debugging and root-cause analysis

Remote Demining Systems Engineering Student, CDS

May 2025 - Aug 2025

- Worked closely with a small team to create a functional prototype of a cost-effective remotely-operated demining vehicle. Independently designed (SolidWorks, KiCad) and built powered mechanical flail, debugged power distribution and control systems (CANbus), and consulted with colleagues to implement software for control over wifi (RPI with router)
 - Presented the system to delegates from various governmental and private defence and military agencies, including a representative working with the UN on demining efforts
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EDUCATION

BASc in Mechatronics and Robotics Engineering

Queen's University at Kingston

Sep 2024 - Apr 2028

4.2 GPA on 4.3 scale

ADDITIONAL INFORMATION

- **Software Skills:** C/C++, Python, Bash, React TSX/JSX, Tailwind CSS, PHP, Git/GitLab, Gerrit, BitBucket, Grafana, Linux systems, UI/UX Agile workflow, TestRail, SwaggerUI, REST APIs
- **Hardware Skills:** SolidWorks, FEA, Technical drawing, Mechanical prototyping, Lab work (oscilloscope, spectrum analyzer, etc.), KiCad, LTSpice, Quartus (Altium), FPGA design, VHDL
- **Languages:** English, French
- **Certifications & Awards:** DELF B2 French Certification, Queen's Engineering Dean's Scholar 2024-2025, FIRST Dean's List Semifinalist Award, Eric R. & Lorraine Davis Memorial Award